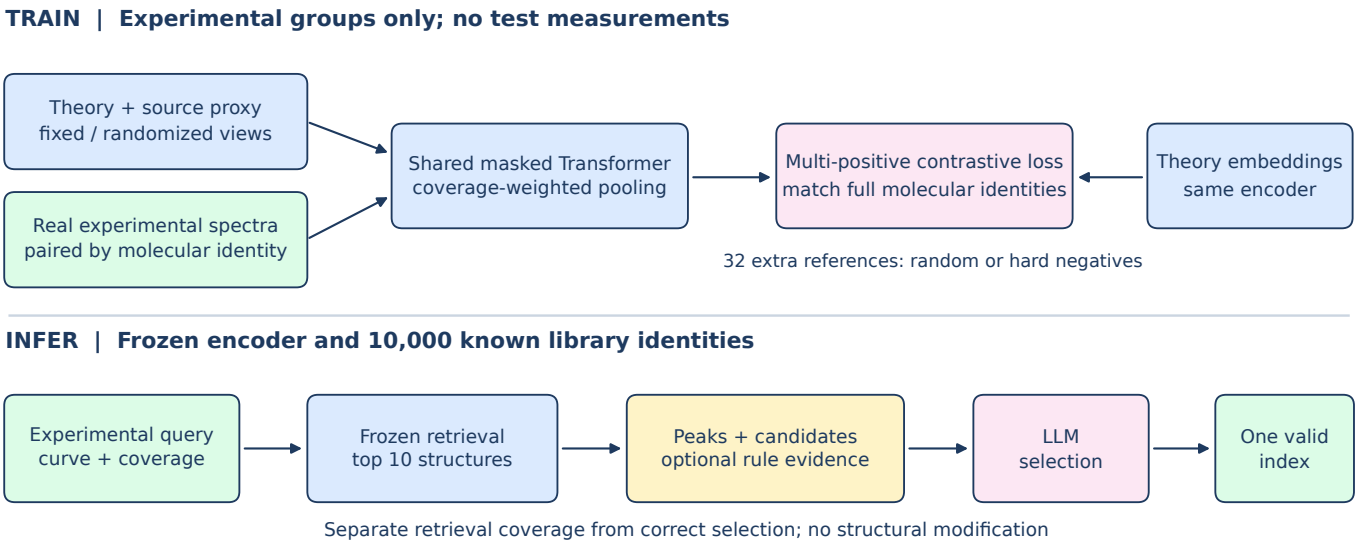
TRAIN | Experimental groups only; no test measurements
Theory + source proxy
fixed / randomized views
Real experimental spectra
paired by molecular identity
Shared masked Transformer
coverage-weighted pooling
Multi-positive contrastive loss
match full molecular identities
Theory embeddings
same encoder
32 extra references: random or hard negatives
INFER | Frozen encoder and 10,000 known library identities
Experimental query
curve + coverage
Frozen retrieval
top 10 structures
Peaks + candidates
optional rule evidence
LLM
selection
One valid
index
Separate retrieval coverage from correct selection; no structural modification